

## **CONCEPT NOTE**

# **A HYBRID ENSEMBLE-BASED MACHINE LEARNING FRAMEWORK FOR ELECTRICITY THEFT DETECTION IN SMART GRIDS UNDER SEVERE CLASS IMBALANCE**

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## **1. INTRODUCTION AND PROBLEM CONTEXT**

The global energy sector is adopting smart grid technology, integrating advanced metering infrastructure with real-time monitoring systems to optimize electricity distribution and consumption. Smart grids enhance operational efficiency, improve grid reliability, and facilitate the integration of renewable energy. According to the International Energy Agency, global investment in smart grid infrastructure reached \$32.8 billion in 2023, with projections indicating a compound annual growth rate of 18.4% through 2030.

However, digitalization has introduced new vulnerabilities, particularly electricity theft and non-technical losses. These fraudulent activities, including meter tampering, unauthorized connections, and billing irregularities, cost the global energy industry approximately \$96 billion annually. In sub-Saharan Africa, technical and non-technical losses average 18%, significantly impacting utility financial sustainability. Kenya's electricity sector exemplifies these challenges, with Kenya Power and Lighting Company (KPLC) reporting system losses of 23.4% in fiscal year 2022/2023, with substantial portions attributed to theft and fraud.

The deployment of smart meters across Kenya, with over 2.1 million advanced metering units installed as of 2024, has generated unprecedented volumes of fine-grained electricity consumption data. However, utilities have yet to fully leverage this data wealth for proactive fraud detection. Machine learning and artificial intelligence techniques have emerged as powerful tools for analyzing vast consumption datasets to identify anomalous patterns indicative of theft.

The State Grid Corporation of China (SGCC) publicly released a benchmark dataset containing daily electricity consumption records of 42,372 customers spanning 1,035 consecutive days from January 1, 2014, to October 31, 2016. This dataset includes labeled instances of 3,615 electricity thieves (8.5%) versus 38,757 honest consumers (91.5%). The significant class imbalance inherent in this dataset mirrors real-world conditions where fraudulent cases constitute a small minority, presenting substantial methodological challenges for traditional machine learning approaches. Despite extensive research utilizing this dataset, existing methods continue to

struggle with achieving high detection rates for theft cases while maintaining acceptable false positive rates.

This research proposes to develop a focused, viable hybrid machine learning framework specifically designed to address the class imbalance challenge through the integration of SMOTE+ENN preprocessing with optimized XGBoost classifier. This approach balances data quality, removes noise, improves recall, reduces false positives, and enhances generalization while maintaining computational feasibility for deployment in real-world smart grid environments.

## **2. PROBLEM STATEMENT: THE CHALLENGE OF ELECTRICITY THEFT AND CLASS IMBALANCE**

Electricity theft represents a pervasive and economically devastating challenge for power utilities globally, with non-technical losses attributable to fraudulent consumption costing the industry approximately \$96 billion annually. In Kenya specifically, Kenya Power reported system losses of 23.4% in 2022/2023, substantially eroding utility revenues, compromising grid reliability, and ultimately imposing higher electricity costs on legitimate consumers. The deployment of smart metering infrastructure has generated vast repositories of granular consumption data that theoretically enable automated detection of anomalous patterns indicative of theft. However, utilities lack effective computational frameworks capable of reliably identifying fraudulent behavior from this data deluge.

The fundamental technical challenge lies in the severe class imbalance characteristic of electricity theft detection problems. In realistic scenarios, fraudulent customers typically represent only 5-15% of the total customer base, creating a highly skewed dataset where theft cases constitute the minority class. The SGCC benchmark dataset exemplifies this challenge, with electricity thieves comprising merely 8.5% (3,615 cases) versus 91.5% (38,757 cases) of honest consumers. This extreme imbalance causes traditional machine learning classifiers to exhibit a bias toward the majority class, resulting in models that achieve deceptively high overall accuracy (often exceeding 90%) by simply predicting the majority class for most instances, while fundamentally failing to detect actual theft cases, the very objective of the detection system.

Empirical evidence demonstrates this problem's severity. A comprehensive analysis of 47 published studies utilizing the SGCC dataset revealed that while average accuracy across approaches exceeded 91%, recall (true positive rate) for detecting theft cases ranged from only 45% to 78%, meaning that 22-55% of actual electricity thieves evaded detection. Furthermore, precision rates varied widely from 38% to 84%, indicating that many models generate excessive false positives, triggering unnecessary field investigations that waste utility resources and potentially alienate honest customers. The cost of missed detections is substantial; each undetected theft case results in continuous revenue loss, estimated at \$1,200-\$3,500 annually per customer in emerging markets, while false positives incur field inspection costs averaging \$150-\$400 per investigation.

Current approaches to addressing class imbalance exhibit significant limitations:

**Data-level limitations:** Standard resampling techniques, such as SMOTE, generate synthetic theft cases but may introduce noise and unrealistic samples that fail to capture sophisticated theft behaviors. Conversely, simple under-sampling discards potentially valuable information about legitimate consumption patterns. The absence of noise removal mechanisms after synthetic sample generation leaves datasets contaminated with borderline cases that confuse classifiers.

**Algorithm-level limitations:** Cost-sensitive learning approaches and threshold adjustments improve recall at the expense of precision, yielding unacceptable false positive rates. Single-model deep learning approaches demonstrate improved feature extraction but remain computationally intensive and lack interpretability, hindering operational deployment.

**Ensemble method limitations:** While ensemble approaches show promise, existing frameworks often employ overly complex heterogeneous combinations without systematic validation of practical viability. The computational overhead of maintaining multiple diverse classifiers creates deployment challenges for resource-constrained utilities.

**Solution viability gap:** Most research focuses on maximizing performance metrics without adequate consideration of computational efficiency, implementation complexity, and operational feasibility. There is a critical need for a focused, practical solution that achieves strong detection performance while remaining computationally viable and interpretable for real-world utility deployment.

This research addresses these gaps by developing a streamlined hybrid framework that integrates SMOTE+ENN preprocessing with optimized XGBoost classification. SMOTE+ENN combines the benefits of synthetic minority over-sampling with intelligent noise removal through Edited Nearest Neighbors, addressing both class imbalance and data quality simultaneously. XGBoost provides state-of-the-art gradient boosting performance with built-in class imbalance handling, computational efficiency, and interpretability through feature importance metrics. This focused approach prioritizes solution viability and practical deployment over theoretical complexity.

### 3. JUSTIFICATION AND IMPORTANCE OF THE RESEARCH

This research addresses a critical socio-economic and technical challenge with far-reaching implications for energy security, utility financial sustainability, and equitable electricity access in developing economies.

#### (a) Why is the problem important?

**Economic Impact:** Electricity theft imposes staggering economic costs. Global annual losses from non-technical losses reach \$96 billion. In Kenya, KPLC's 23.4% system losses in 2022/2023 translated to approximately KSh 28.7 billion (approximately \$220 million) in unrealized revenue, equivalent to the annual electricity needs of over 500,000 households. Each 1% reduction in losses could fund electrification for 21,400 additional households annually.

**Operational Sustainability:** Revenue losses reduce capital available for network maintenance and expansion, creating a vicious cycle where deteriorating infrastructure increases technical losses and service disruptions. Persistent high losses have necessitated repeated government bailouts for Kenya Power, diverting public resources from critical development priorities.

**Tariff Equity:** Honest consumers subsidize fraudulent consumption through higher tariffs, paying 10-40% more than in a theft-free system. This regressive burden disproportionately affects low-income households, potentially pushing vulnerable populations into energy poverty.

**Grid Reliability and Safety:** Electricity theft compromises grid reliability and safety through unaccounted loads that distort demand forecasting and load balancing, potentially triggering localized overloads and blackouts. Illegal connections pose serious fire and electrocution hazards.

**(b) Can the selection of the proposed research topic be justified?**

**Technical Relevance:** Kenya's deployment of 2.1 million smart meters and similar initiatives across Africa generate massive consumption datasets that remain grossly underutilized. This research directly addresses the need for practical analytical frameworks that transform smart meters from passive revenue collection devices into active security tools.

**Methodological Innovation with Practical Viability:** Unlike overly complex approaches, this research proposes a focused, implementable solution. SMOTE+ENN addresses class imbalance while maintaining data quality through intelligent noise removal. The technique balances data distribution, removes borderline cases that introduce confusion, improves minority class recall, reduces false positives through cleaner decision boundaries, and enhances model generalization. XGBoost provides state-of-the-art performance with computational efficiency suitable for utility deployment, built-in mechanisms for handling class imbalance through the `scale_pos_weight` parameter, and interpretability through feature importance ranking. This streamlined approach prioritizes solution viability, a critical requirement highlighted by utility stakeholders.

**Practical Applicability:** The framework emphasizes operational deployment feasibility with manageable computational requirements, clear implementation procedures, and interpretability suitable for utilities with limited data science expertise. By focusing on a proven, efficient methodology rather than theoretical complexity, this research enables replication by resource-constrained utilities.

**Developing Economy Context:** This research specifically targets challenges faced by utilities in developing economies where electricity theft rates are highest, loss reduction imperatives are most urgent, and technical capacity is most constrained. The framework's computational efficiency and implementation simplicity maximize developmental impact.

**Policy Relevance:** Effective electricity theft detection directly supports Sustainable Development Goals, including SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), and SDG 11 (Sustainable Cities and Communities). Improved

loss reduction contributes to Kenya Vision 2030 objectives of providing reliable, affordable energy to support industrialization.

### **(c) Can others be convinced of its importance?**

**Utility Stakeholders:** A focused, implementable solution addresses utilities' practical constraints. Demonstrating that a streamlined machine learning approach can significantly improve theft detection while remaining computationally efficient presents a compelling value proposition. Quantifying potential revenue recovery, even a conservative 2% reduction in Kenya Power's losses would recover KSh 2.4 billion annually, providing concrete justification for investment.

**Regulatory Bodies:** Energy regulators prioritize loss reduction as a key performance indicator. This research provides evidence-based insights into effective, practical technological approaches for combating theft, potentially informing regulatory frameworks around smart metering requirements and performance incentives.

**Academic Community:** The focused methodology addresses class imbalance challenges applicable across numerous domains, including credit card fraud, network intrusion detection, and medical diagnosis. The availability of the public SGCC dataset facilitates reproducible research and comparative evaluation.

**Development Partners:** International development organizations recognize energy sector sustainability as foundational for economic development. Demonstrating cost-effective, implementable technological solutions for reducing electricity losses addresses a critical barrier to utility financial viability and energy access expansion.

## **4. RESEARCH OBJECTIVES**

### **Main Objective**

The main objective of this research is to develop and validate a practical hybrid machine learning framework for electricity theft detection in smart grids that effectively addresses severe class imbalance through SMOTE+ENN preprocessing integrated with optimized XGBoost classification, achieving superior detection performance compared to existing approaches while



maintaining computational efficiency, interpretability, and operational feasibility for deployment in resource-constrained utility environments.

### **Specific Objectives**

**Objective 1: To design and implement a SMOTE+ENN preprocessing pipeline optimized for electricity consumption data that balances class distribution while removing noisy samples to enhance classification performance.**

*Measurable outcomes:* Development of a complete preprocessing pipeline with documented parameter configurations; quantitative evaluation of preprocessing effectiveness measured by improved class separability metrics and reduced noise levels; generation of balanced training datasets validated through distribution analysis; extraction of discriminative temporal and statistical features capturing consumption patterns.

**Objective 2: To develop and optimize an XGBoost-based classification model with hyperparameter tuning strategies that maximize theft detection recall while controlling false positive rates, incorporating mechanisms to address overfitting and computational efficiency concerns.**

*Measurable outcomes:* Implementation of XGBoost classifier with documented hyperparameter optimization procedures; achievement of minority class (theft) recall exceeding 85% while maintaining precision above 75% on SGCC test set; demonstration of computational efficiency suitable for utility deployment; validation of overfitting mitigation through cross-validation and learning curve analysis; generation of interpretable feature importance rankings.

**Objective 3: To conduct comprehensive comparative evaluation of the proposed framework against existing state-of-the-art electricity theft detection methods using the SGCC benchmark dataset, analyzing performance, computational efficiency, and interpretability dimensions.**

*Measurable outcomes:* Systematic reproduction and evaluation of at least 3 existing state-of-the-art approaches; comparative performance analysis across multiple evaluation metrics (accuracy, recall, precision, F1-score, AUC-ROC, G-mean); computational efficiency benchmarking,

measuring training time and inference latency; interpretability assessment through feature importance analysis; statistical validation of performance improvements.

**Objective 4: To develop practical deployment guidelines with implementation procedures, performance monitoring protocols, and continuous improvement mechanisms suitable for adoption by resource-constrained electricity distribution utilities.**

*Measurable outcomes:* Creation of comprehensive technical documentation including system architecture specifications and deployment procedures; development of performance monitoring and model retraining protocols addressing concept drift; cost-benefit analysis quantifying potential revenue recovery and implementation requirements; delivery of training materials for utility personnel.

## **5. LITERATURE REVIEW AND IDENTIFIED GAPS**

Electricity theft detection has evolved from manual inspection methods to sophisticated machine learning approaches. Early applications employed classical algorithms, including Support Vector Machines, Decision Trees, and k-Nearest Neighbors. Nagi et al. (2011) pioneered SVM-based theft detection, achieving 85% accuracy but suffering from high false positive rates exceeding 30%.

The release of the SGCC benchmark dataset in 2017 catalyzed intensive research into advanced detection methods. Zheng et al. (2018) introduced Wide and Deep Convolutional Neural Networks, achieving 89.5% accuracy and 75.8% recall. Li et al. (2019) applied LSTM networks to model temporal dependencies, while Gao et al. (2020) proposed attention mechanisms achieving 82.3% recall and 78.1% precision. Despite improvements, deep learning approaches remained computationally intensive and lacked interpretability.

Recognizing class imbalance as the central challenge, researchers explored various mitigation strategies. Punmiya and Choe (2019) combined SMOTE with deep neural networks, improving theft recall from 68% to 79%. However, standard SMOTE exhibited limitations in high-dimensional space and failed to account for noise in synthetic samples. Khan et al. (2020)

implemented cost-sensitive Random Forests achieving 81% recall but suffering increased false positive rates approaching 25%.

Gradient boosting methods, particularly XGBoost, have demonstrated strong performance across imbalanced classification tasks. XGBoost's built-in `scale_pos_weight` parameter enables explicit class imbalance handling, while its regularization mechanisms prevent overfitting. Chen and Guestrin (2016) demonstrated XGBoost's superiority over traditional methods across multiple benchmarks. However, XGBoost's application to electricity theft detection has been limited, with most research not leveraging its full potential through systematic hyperparameter optimization and integration with intelligent preprocessing.

SMOTE+ENN combines synthetic minority over-sampling with Edited Nearest Neighbors to simultaneously address class imbalance and remove noisy samples. ENN identifies and removes instances whose class label differs from the majority of their k-nearest neighbors, cleaning decision boundaries and reducing false positives. Batista et al. (2004) demonstrated that SMOTE+ENN outperforms standard SMOTE across multiple imbalanced datasets by improving both recall and precision. However, SMOTE+ENN's application to electricity theft detection remains underexplored.

### **Technical Gaps Identified**

**Gap 1: Insufficient Focus on Solution Viability** Most research prioritizes maximizing performance metrics through complex architectures without adequate consideration of implementation complexity, computational requirements, and operational feasibility. Heterogeneous ensembles with multiple diverse classifiers create deployment challenges for resource-constrained utilities lacking GPU infrastructure and specialized expertise.

**Gap 2: Inadequate Integration of Preprocessing and Classification** Existing approaches treat preprocessing and classification as separate stages without systematic optimization of their interaction. SMOTE+ENN's noise removal capability combined with XGBoost's gradient boosting strength represents an underexplored integration that addresses both data quality and classification performance simultaneously.

**Gap 3: Limited Attention to Computational Efficiency** Deep learning approaches achieve strong performance but require substantial computational resources unsuitable for many utility

environments. The need for efficient, deployable solutions that balance performance with practical constraints remains unaddressed.

**Gap 4: Insufficient Overfitting Mitigation Strategies** While sophisticated preprocessing improves data quality, potential overfitting risks require systematic mitigation through regularization, cross-validation, and careful hyperparameter tuning. Current literature provides limited guidance on managing this trade-off in electricity theft detection contexts.

This research specifically addresses these gaps by developing a streamlined, practical framework that systematically integrates SMOTE+ENN preprocessing with optimized XGBoost classification, emphasizing solution viability, computational efficiency, and overfitting mitigation while achieving state-of-the-art detection performance.

## 6. METHODOLOGY OVERVIEW

This research employs a systematic experimental methodology using the State Grid Corporation of China (SGCC) electricity theft detection dataset. The methodology is structured into five focused phases emphasizing practical implementation and solution viability.

**Dataset:** The SGCC dataset contains daily electricity consumption records for 42,372 customers over 1,035 consecutive days (January 1, 2014 to October 31, 2016), with 3,615 labeled theft cases (8.5%) and 38,757 honest consumers (91.5%).

### Phase 1: Data Acquisition and Exploratory Analysis

**Objective Alignment:** Supports all objectives by establishing foundational data understanding.

#### Methods:

1. **Data Loading and Quality Assessment:** Import SGCC dataset and conduct comprehensive quality assessment identifying missing values, outliers, and zero-consumption sequences.
2. **Missing Value Treatment:** Implement linear interpolation for isolated gaps ( $\leq 3$  days) and forward-fill/backward-fill for extended sequences.

3. **Exploratory Data Analysis:** Generate descriptive statistics for honest and theft groups, visualize consumption distributions and temporal patterns, conduct statistical hypothesis testing to assess class differences.
4. **Class Distribution Analysis:** Quantify imbalance ratio (11:1), analyze temporal distribution of theft cases.

**Tools:** Python 3.9+, pandas, NumPy, matplotlib, seaborn, scipy

## **Phase 2: Feature Engineering and SMOTE+ENN Preprocessing**

**Objective Alignment:** Directly addresses Specific Objective 1.

### **Feature Engineering:**

Extract comprehensive feature set capturing consumption characteristics:

- **Statistical features (8):** Mean, median, standard deviation, coefficient of variation, minimum, maximum, range, skewness
- **Trend features (6):** Linear trend slope over various periods, trend direction changes, acceleration
- **Temporal patterns (7):** Weekday vs. weekend ratio, peak consumption patterns, day-to-day variability, seasonal indicators
- **Anomaly indicators (4):** Zero-consumption days, sudden consumption changes, volatility index

**Total:** 25+ engineered features

**Feature Normalization:** Apply Min-Max scaling to ensure features lie within [0,1] range.

**Tools:** Python scikit-learn, imbalanced-learn

## **Phase 3: XGBoost Model Development and Optimization**

**Objective Alignment:** Directly addresses Specific Objective 2.

### **XGBoost Configuration and Hyperparameter Optimization:**

**Base Configuration:**

- Objective: binary: logistic (binary classification)
- Eval\_metric: AUC, Recall, Precision
- scale\_pos\_weight: Set based on class imbalance ratio to penalize minority class misclassification

**Optimization Strategy:** Implement Grid Search or Bayesian Optimization to systematically explore the hyperparameter space. Define the objective function as a weighted combination prioritizing recall (60%), precision (25%), and F1-score (15%) on the validation set. Conduct 5-fold stratified cross-validation to ensure robust parameter selection.

**Rationale:** XGBoost provides state-of-the-art gradient boosting with built-in class imbalance handling, computational efficiency, and interpretability. Its regularization mechanisms directly address overfitting concerns. Parameter tuning balances the bias-variance tradeoff, optimizing generalization while maintaining strong detection performance.

**Model Training:** Train an optimized XGBoost model on the SMOTE+ENN preprocessed training set. Evaluate on the validation set to confirm hyperparameter effectiveness. Reserve test set exclusively for final performance evaluation.

**Tools:** Python XGBoost, scikit-learn, scikit-optimize

**Phase 4: Comprehensive Comparative Evaluation**

**Objective Alignment:** Directly addresses Specific Objective 3.

**Benchmark Approaches:** Reproduce and evaluate at least 3 existing state-of-the-art methods:

1. Standard XGBoost without SMOTE+ENN (baseline)
2. Random Forest with SMOTE
3. Wide and Deep CNN (Zheng et al., 2018)
4. LSTM with class weights (if computationally feasible)

## Evaluation Metrics:

Performance metrics addressing class imbalance:

1. **Recall (Sensitivity):**  $TP / (TP + FN)$  - measures theft detection rate
2. **Precision:**  $TP / (TP + FP)$  - measures false positive control
3. **F1-Score:** Harmonic mean of precision and recall
4. **G-Mean:**  $\sqrt{(\text{Recall} \times \text{Specificity})}$  - balanced metric for imbalanced data
5. **AUC-ROC:** Area under ROC curve
6. **Matthews Correlation Coefficient (MCC):** Robust metric accounting for imbalance
7. **Accuracy:** Overall correctness (reported but not primary metric)

**Statistical Validation:** Conduct 10-fold stratified cross-validation for robust performance estimation. Apply paired t-tests to assess statistical significance ( $p < 0.05$ ) of performance differences.

## Computational Efficiency Analysis:

- Training time (wall-clock time)
- Inference latency (prediction time per sample)
- Memory requirements
- Hardware requirements

**Interpretability Assessment:** Generate XGBoost feature importance rankings using:

- Gain: Average gain across all splits using the feature
- Weight: Number of times feature appears in trees
- Cover: Average coverage of splits using the feature

Visualize top 20 most important features. Analyze feature patterns distinguishing theft from honest consumption.

**Error Analysis:** Examine false positives and false negatives to identify systematic error patterns and potential improvement directions.

**Tools:** Python scikit-learn, matplotlib, seaborn, SciPy

## **Phase 5: Operational Deployment Framework Development**

**Objective Alignment:** Directly addresses Specific Objective 4.

### **Methods:**

1. **System Architecture Design:** Document framework components, data flows, integration points with utility systems, and hardware/software requirements.
2. **Deployment Procedures:** Create a step-by-step deployment guide including environment setup, dependency installation, model loading, and batch prediction workflows.
3. **Performance Monitoring Protocols:** Define KPIs (detection rate, false positive rate, investigation success rate) and monitoring frequency.
4. **Model Retraining Procedures:** Develop protocols for periodic retraining incorporating newly labeled data from field investigations. Define retraining triggers and update procedures.
5. **Cost-Benefit Analysis:** Quantify potential revenue recovery, false positive investigation costs, implementation costs, and ROI for a representative utility scenario.
6. **Training Materials:** Develop user guides and technical documentation for utility personnel.

**Tools:** Python documentation platforms, presentation software



## 7. EXPECTED OUTPUTS AND IMPACT

### Expected Outputs

1. **Practical Hybrid Machine Learning Framework:** A validated, implementable framework integrating SMOTE+ENN with optimized XGBoost, delivered as documented Python software with implementation guidelines.
2. **Master's Thesis Document:** Comprehensive thesis (80-120 pages) documenting literature review, methodology, experimental results, and conclusions.
3. **Peer-Reviewed Publications:** At least 1-2 journal manuscripts:
  - "Hybrid SMOTE+ENN and XGBoost Framework for Electricity Theft Detection Under Severe Class Imbalance" (target: IEEE Access, Applied Energy, or similar)
4. **Open-Source Repository:** Complete GitHub repository with preprocessing pipeline, XGBoost implementation, evaluation scripts, documentation, and example notebooks.
5. **Operational Deployment Package:** Documentation including system architecture, deployment procedures, monitoring protocols, cost-benefit analysis, and training materials.
6. **Benchmark Results:** Performance comparison establishing new baseline for SGCC dataset using focused, viable methodology.

### Expected Impact

**Methodological Impact:** The research will demonstrate that focused, practical methodologies can achieve state-of-the-art performance while maintaining implementation viability.

SMOTE+ENN integrated with optimized XGBoost addresses class imbalance through intelligent noise removal and efficient gradient boosting, advancing practical machine learning for imbalanced classification problems.

**Practical Utility Impact:** Adoption by electricity utilities will enable significant revenue recovery. Conservative estimates suggest achieving 85% theft detection recall could reduce non-technical losses by 1.5-2.5 percentage points. For a utility serving 500,000 customers with average monthly revenue of \$25 and 20% system losses (half from theft), a 2% loss reduction translates to annual revenue recovery of approximately \$3 million. Reduced false positive rates (below 10%) minimize unnecessary investigations, conserving operational resources. Enhanced interpretability through XGBoost feature importance improves investigator efficiency.

**Computational Feasibility:** XGBoost's efficiency enables deployment on standard utility computing infrastructure without requiring GPU acceleration, reducing implementation barriers for resource-constrained utilities.

**Developmental Impact:** Improved theft detection contributes to electricity sector financial sustainability, enabling infrastructure investment and service expansion. Revenue recovery can fund electrification programs. Reduced losses lower electricity costs for consumers.

**Policy and Regulatory Impact:** Research findings inform energy regulatory policy regarding smart metering, data analytics, and loss reduction requirements. Quantified cost-benefit analysis supports regulatory approval for utility analytics investments.

**Academic Impact:** Publication in peer-reviewed venues contributes to discourse on practical machine learning for imbalanced classification. Open-source release facilitates reproducible research and comparative evaluation.

**Technology Transfer:** The validated framework presents opportunities for integration into smart metering platforms, enabling technology vendors to offer practical fraud detection solutions to developing economy utilities.

**Addressing Identified Challenges:** The research systematically addresses overfitting through XGBoost regularization parameters (`gamma`, `reg_alpha`, `reg_lambda`), early stopping, and cross-validation. Computational costs remain manageable through XGBoost's efficiency and focus on single optimized model rather than complex ensemble. Parameter tuning strategies provide clear guidance for balancing performance and generalization.

In summary, this research delivers a practical, validated electricity theft detection framework that advances both methodological understanding and operational utility capabilities, with measurable impacts on revenue recovery, system efficiency, and sustainable development particularly relevant to resource-constrained developing economies.

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## **APPENDIX: DATASET INFORMATION**

**Dataset Name:** SGCC Electricity Theft Detection Dataset

**Source:** State Grid Corporation of China (available on Kaggle)

**URL:** <https://www.kaggle.com/datasets/bensalem14/sgcc-dataset>

### **Dataset Characteristics:**

- Total Customers: 42,372
- Time Period: 1,035 consecutive days (January 1, 2014 - October 31, 2016)
- Theft Cases: 3,615 (8.5%)
- Honest Consumers: 38,757 (91.5%)
- Class Imbalance Ratio: 1:11
- Data Format: Daily electricity consumption values (kWh)
- Labels: Binary (0=honest, 1=theft)